Coal's Decline Imposed Large, Lasting Costs on Workers

Lessons from "Transitional Costs and the Decline of Coal: Worker-Level Evidence" by Jonathan Colmer, Eleanor Krause, Eva Lyubich & John Voorheis, Working Paper, 2026.

When U.S. coal production and employment suddenly fell by half in the 2010s, the workers it left behind never caught back up. The most coal-dependent workers lost close to two years' worth of earnings over the following decade — and neither switching industries nor moving to a new region did much to help recoup those losses.

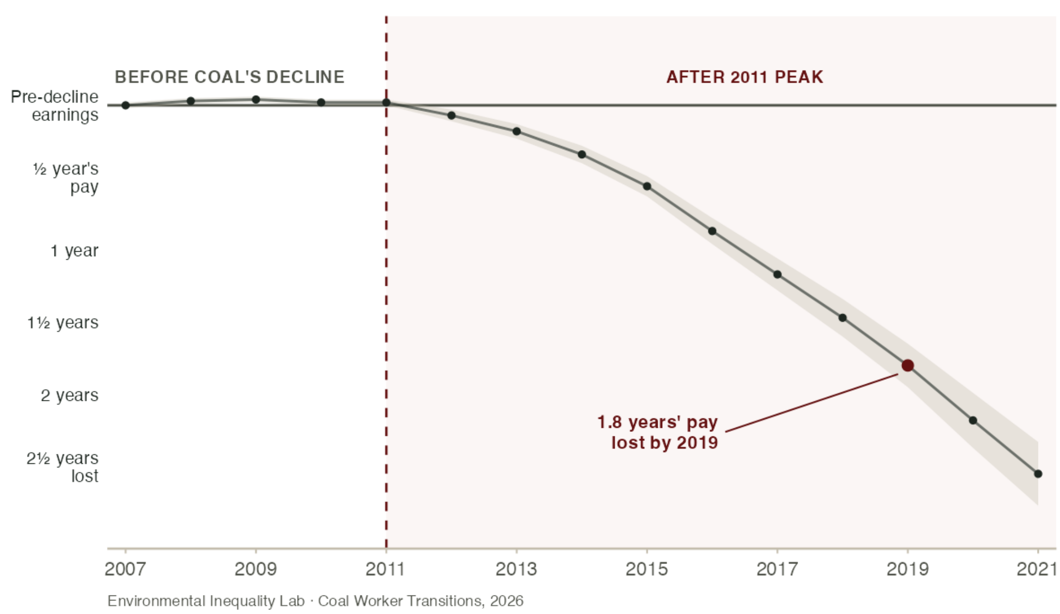
BACKGROUND

U.S. coal mining employment has fallen more than half since its peak in 2011, as cheap natural gas from the fracking boom and falling renewable energy costs pushed coal out of the power sector. Earlier research shows that coal *communities* lost jobs, earnings, and tax revenue. However, aggregate numbers may hide how individual workers fared. Some may have quickly moved to new jobs, while others may have borne heavy **transitional costs**: the lasting losses to earnings and employment that workers bear when an industry shrinks. Colmer et al. wanted to use the decline of coal to understand individual-level transitional costs, and how that might inform the broader question of what the energy transition could mean for workers in other high-wage, regionally concentrated industries.

To assess the private costs of coal's decline and understand if aggregated patterns held for individual workers, Colmer et al. assembled administrative records covering nearly every person who worked in U.S. coal mining before, during, and after coal's 2011 peak. This let them follow each worker's earnings, employment, and location year by year, even if workers worked outside coal or were unemployed. They could then compare those paths with otherwise similar workers who depended less on coal for their livelihoods. **This comparison showed that coal-dependent workers experienced persistent losses in earnings and employment following coal's decline.**

COAL WORKERS' EARNINGS FELL FURTHER BEHIND COMPARABLE WORKERS EVERY YEAR

Source: recreated from Colmer et al. (2026), Fig. 7a.



Note: Each point is the cumulative earnings gap between workers whose livelihoods depended most on coal and otherwise similar workers who depended on it less, measured in years of pre-decline pay. After coal's 2011 peak, more- and less-coal-dependent workers' earnings steadily diverge — by 2019 the gap reaches 1.8 years' worth of earnings. Values digitized from the published figure.

HOW THE STUDY TESTED IT

Measuring what coal's decline cost workers is challenging, because a career is shaped by many forces at once. One might simply track coal workers' earnings after 2011 and call the drop the cost — but aging, the broader economy, and ordinary job churn were all pulling on those same paychecks, so a raw decline can't be pinned on coal alone. The next step is to compare coal workers against everyone else. Yet coal workers tend to be better paid, clustered in a few isolated regions, and different from other workers in age and education, so any earnings gap between them could reflect those existing differences rather than what coal's decline did to them.

Colmer et al. narrowed the comparison to workers who looked alike on those traits — evening out age, education, earnings, and location — and set the most coal-dependent workers beside otherwise similar workers who depended on coal far less. This holds only if the two groups would have followed the same earnings path absent the industry's collapse, which the researchers have to assume. That assumption is easier to credit here because the collapse was driven by national energy prices rather than any one worker's choices, so the most coal-dependent workers were hit by a force outside their control. If coal's decline had left workers unharmed, or affected both groups equally, no earnings gap should have opened between them.

WHAT THE STUDY FOUND

This is not what the researchers found. Between 2012 and 2019, **the most coal-dependent workers lost cumulative earnings equal to roughly 1.8 times a full pre-decline year of pay**. The losses came through two channels at once: compared with similar workers, coal-dependent workers spent about five more months out of work *and*, in the years they did work, earned roughly 19 percent less.

Workers couldn't simply move or retrain their way out. Coal paid well — about $13,000 a year more than comparable work — and its jobs are concentrated in a few isolated regions where entire economies lean on the industry, so nearby alternatives were scarce. **Those who left coal for other industries earned over 30 percent less than they had in coal, and relocating to a new labor market barely helped.** Where workers displaced by factory closings often find their way into new jobs, a large share of coal workers stopped working altogether, a sign their skills didn't transfer to the jobs around them.

As earnings fell, coal-dependent workers leaned on government transfers, especially Social Security Disability Insurance (SSDI), and drew from retirement savings earlier. These payments cushioned the blow but replaced only a fraction of what workers lost. **A decade on, coal's decline had carved a deep and persistent hole in workers' earnings that neither the labor market nor the safety net filled.**

Coal's decline is an early look at what shrinking demand for fossil fuels could mean for workers in regionally concentrated, high-wage industries. Coal is just one industry, and its geography is unusually isolating, so its exact path won't repeat everywhere. What the study establishes is narrower and still sobering: **when a high-wage, regionally concentrated industry contracts, its workers can bear the cost for a decade, and existing safety nets catch only part of the fall.**

FOR MORE INFORMATION

Colmer, Jonathan, Eleanor Krause, Eva Lyubich, and John Voorheis. 2026. "Transitional Costs and the Decline of Coal: Worker-Level Evidence." Working paper. researchonline.lse.ac.uk/id/eprint/126795.

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